

## 3.8 Differential equations

### Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

Separable differential equations rearrange to integrate functions of  $y$  and  $x$  on opposite sides. Model assumptions determine meaningful domain; initial conditions select a particular solution. Exponential change arises when rate is proportional to current amount.

#### Worked example: follow the reasoning

$\frac{dy}{dx}=2y$ ,  $y(0)=3$  yields  $\ln(y)=2x+C$  and  $y=3e^{2x}$ .

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 Solve  $\frac{dy}{dx}=-0.5y$  with  $y(0)=8$ .

[3]

Final answer: \_\_\_\_\_

## Answers and explanation

### Question 1 [3 marks]

Separate  $dy/y = -0.5 dx$  [1];  $\ln abs(y) = -0.5x + C$  [1];  $y = 8e^{(-0.5x)}$  [1].

### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.